

Wrong Family, Right Fit

High-Fidelity Parameter Fitting for Spatial Point Processes Near Complete Spatial Randomness

AISTATS submission plan

Abstract [DRAFT]

The quality of an SPP fit is typically assessed by distance between inferred and true generating process parameters. Existing methods - such as minimum contrast - require prior knowledge of the underlying family, and their performance degrades near complete spatial randomness (CSR): despite the generating process not being Poisson, its second-order structure (e.g. K -function) is almost indistinguishable from Poisson's. This is an inherent property of the problem, not a failure of methodology.

Instead, we evaluate a fit by generating simulations from the predicted parameters and scoring their fidelity to an independent held-out replicate, using a strictly proper scoring rule on local configurations, normalized between a Poisson (CSR) model (0) and the true model (1).

We propose a two-stage pipeline: a classifier determines the family, followed by family-specific parameter estimators. The first stage acts as a filter, routing near-CSR to the Poisson estimator.

Near CSR, patterns become inherently difficult to classify: a diverse set of classifiers (gradient-boosted trees, linear models, neural networks) over two feature types (summary functions, topological features) converge to similar accuracy, which degrades as a family's parameters approach the CSR limit.

Reproduction fidelity is 0.92, despite the classifier only achieving 53% accuracy.

The inclusion of topological features never hurts parameter estimation, and helps substantially where second-order statistics are blind by construction: for the cell process they cut estimation error by 15% and raise fidelity from 0.92 to 1.00.

Numbers of record (all on the fixed test split, θ -bootstrap intervals). Accuracy: `oneshot/results/default/compare/summary.md`. Skill 0.93: `evaluation/heldout` (480 clouds, stratified by regime). Wrong-family skill: `evaluation/heldout_ph.ablation` (1080 clouds in the strong-signal strata; 357 misassigned). Cell error $0.142 \rightarrow 0.120$ (classical+PH) and 0.117 (all six filtrations).

The Story [INFORMAL]

- All models struggle with classification/inference on clouds near CSR. Unless you cherry-pick your prior (bad), you will have to deal with a large number of near-CSR clouds. Bad performance on these near-CSR clouds is not a failure of the models, it's an inherent property of the problem.
- The pitch: we show that, despite this difficulty, classifiers successfully tell us which generating process is "good enough" for inference.
- Classification accuracy on the 8 processes seems weak (0.53), but the final fidelity score - i.e. how well the inferred parameters reproduce the pattern - is high (0.93).
- We show that when near-CSR but non-Poisson clouds get routed to Poisson inference by the classifier, the result is still high-fidelity.

Contributions

1. **Scoring.** We evaluate a fit by whether clouds generated from it look like the input data.
 - Rescale pattern to unit intensity. Around each of its points, take the neighbours within $R = 1.5$ mean spacings, giving X configurations. . .
 - Embed each configuration as a smoothed density on an $N \times N$ grid with a Gaussian kernel k whose width is fixed by the observed pattern y ,

$$S(P, y) = \frac{1}{2} \mathbb{E} k(\Phi, \Phi') - \frac{1}{m} \sum_i \mathbb{E} k(\phi_i, \Phi),$$

where ϕ_i are y 's configurations and Φ, Φ' are configurations from independent simulations of P .

- S is compared with two reference models on the same pattern: the oracle P^* (true family and θ) and CSR P_0 . Then

$$\text{skill} = \frac{\sum_j [S(P_0, y_j) - S(\hat{P}_j, y_j)]}{\sum_j [S(P_0, y_j) - S(P_j^*, y_j)]},$$

so 0 means no better than CSR and 1 means as good as the truth.

- $\hat{P}_j$ is fitted to x_j and scored on an independent replicate y_j of the same process.
2. **The CSR threshold.** Across 8 families and 11 classifiers + 10 estimators, classification and parameter error collapse at the same boundary, described by one physical coordinate per family (cluster overlap ω , signal-to-noise ζ) times a power of $\bar{n}$. The proper score confirms the boundary independently of any learned model: there, even the true model does not beat CSR.
 3. **Pipeline.** $p(f, \theta | x) = p(f | x) p(\theta | x, f)$: one classifier (its $p(\text{poisson} | x)$ is the CSR filter), then per-family estimators. The flat classifier beats a hierarchical cascade (group, then family, with a reject class) by 5.4 points.
 4. **Where topology helps.** Adding PH to classical summaries never makes estimates significantly worse and helps most where second-order statistics cannot see the structure (cell process: $K(r) = \pi r^2$ exactly; ring process: loops in H_1).

Paper outline (8 pages; check the CFP for the current limit)

section	pages	content, key display
1 Introduction	1.0	Hook: accuracy 0.53 vs skill 0.93. Why accuracy is the wrong target near CSR. Contributions. Fig. 1 (the CSR wall, below).
2 Setting and score	1.25	Families table (8 families, parameters, regime coordinate). Bank: 50 000 $\theta \times 2$ replicates per family, $\bar{n} \in [75, 950]$, split by θ . Kernel score, anchors, skill; Dawid–Sebastiani as secondary. Tab. 1 power check (d' at strongest structure: kernel 1.06–1.30, Poisson floor 0.12; W_1 energy 0.00–0.86).
3 Method	1.0	Factorisation; classifier as filter; per-family estimators on $\log \theta$. Inputs: 111 classical summary-function samples (L, F, G, J), 105 PH summaries per filtration, PersLay, persistence images. Learners: gradient-boosted trees, linear, CNNs. Poisson: $\hat{n} = n$, the exact MLE (all 22 network inputs reach the same $1/\sqrt{\bar{n}}$ floor, so no model is needed).
4 The CSR wall	1.5	Regime coordinates and their fit; per-model detection curves; parameter error vs coordinate; oracle-vs-CSR gain vs coordinate. Structured patterns routed to Poisson have gain $+5 \times 10^{-5} \pm 5 \times 10^{-5}$: calling them Poisson costs nothing.
5 Fidelity	1.5	Tab. 2 skill by family and stratum. Wrong family, right pattern (mis-assigned vs identified). Ablation: flat vs hierarchical. [classical baseline row] [real-data demo]
6 When topology helps	0.75	Paired estimator table; cell and ring; end-to-end ablation.
7 Negative results, limitations	0.5	Training filter; W_1 ; point estimates rather than posteriors; synthetic bank.
Related work	0.5	Minimum contrast / Palm / composite likelihood; global envelope tests; simulation-based inference and amortized model comparison; PH for point processes; neural summaries (Vihrs et al.).

Fig. 1, the money figure. Three panels, each against the family’s regime coordinate u , one curve per model: (a) detection rate (not called Poisson) for all 11 classifiers; (b) estimator error for all 10 estimators; (c) score gain of the true model over CSR. All three collapse at the same u , and the model curves overlap. [to be made; the per-model tables are consistent with it, but panels (a) and (b) have not been plotted per model]

Appendix. Full classifier/estimator tables and paired differences; the 132-configuration network study (`docs/05_results.tex`); regime-coordinate selection; score settings and the settings rejected; pilot of the three added families; training-filter study with the estimated- θ gate.

Evidence behind each claim

claim	number	status
Top classifiers tie	best four 0.533–0.535; all 11 in [0.488, 0.535]	done
Accuracy is at a ceiling	all 11 combined (logistic stacking, fit on val): 0.536 vs best single 0.535	done
Errors are CSR or near-equivalences	merge Matérn I/II: 0.535 $\rightarrow$ 0.62; also nested, ring, LGCP into Thomas: 0.79. Called Poisson: Thomas 35%, ring 26%, LGCP 24%	done
Low Thomas recall is a trade, not a defect	parsimony prior: Thomas recall 0.17 $\rightarrow$ 0.32, accuracy 0.535 $\rightarrow$ 0.500, nested recall 0.55 $\rightarrow$ 0.23	done
Thomas error sits on the κ - μ ridge	κ, μ : 0.79 each; $\bar{n} = \kappa\mu$: 0.20–0.30; $\omega = \sigma\sqrt{\kappa}$: 0.49; σ : 0.54	done
Accuracy rises in regime	best 0.535 all $\rightarrow$ 0.652 at $\tau = 0.9$	done
Fidelity	kernel skill 0.93 (best) / 0.91 (classical), held-out replicate	done
Wrong family, right pattern	misassigned 0.95 vs identified 0.94 (1080 strong-signal clouds); 0.74 vs 0.96 over all strata	done
Routing to Poisson is free	gain $+5 \times 10^{-5} \pm 5 \times 10^{-5}$ on 102 misrouted structured clouds	done
Flat beats hierarchical	-0.054 [$-0.062, -0.046$] family accuracy	done (7 families, pilot)
Regime coordinate explains detection	explained 0.965–0.992 (original four families)	done; [refit for 8]
Score has power	d' table above	done; [write-up in cascade/scoring/scores.tex]
PH never hurts estimates	all 7 paired intervals ≤ 0 or containing 0	done
PH helps where K is blind	cell 0.142 $\rightarrow$ 0.120 / 0.117; ring 0.648 $\rightarrow$ 0.642 (0.518 $\rightarrow$ 0.498 at $\tau = 0.9$)	done
PH end to end	skill 0.94 with and without; cell 0.87 $\rightarrow$ 0.96, interval includes 0	null overall; [cell-only run, 1000 clouds, evaluation/heldout_ph_cell]
Training filter	in-regime gain up to 0.055; estimated- θ gate: no gain	done; [archive script]
Beats classical estimation	—	[missing]
Works on real data	—	[missing]

Presenting the weak numbers

None of these can be fixed by more modelling (the ceiling row above); all can be presented honestly and to our advantage. Put them in the main text next to their explanation, never only in the appendix.

- **Accuracy 0.53.** Report it against the regime coordinate, not only pooled: the pooled value depends on how much prior mass sits near CSR, which is our design choice (0.65 in the strong-signal regime).

Add an equivalence-aware accuracy that merges families which coincide in part of their parameter space, justified by the fidelity of misassigned patterns (skill 0.95).

- **Thomas recall 0.17.** Thomas is the limit of three other families (nested with vanishing meta-clusters, ring with large jitter, and LGCP at second order). Report the parsimony operating point as the demonstration that recall only moves between families.
- **Matérn II recall 0.34** (57% called Matérn I). At low packing both are hard cores of the same radius; Matérn II fidelity is 1.00.
- **Thomas estimator error 0.70.** Report the identifiable combinations $(\bar{n}, \omega, \sigma)$ beside κ and μ ; the κ - μ ridge motivates posteriors as future work. In regime: 0.45.
- **Wrong-family skill 0.74 over all strata.** Skill is unstable where the true model barely beats CSR; quote it only where the gain is resolved (0.95).
- **PH end to end.** Fidelity is saturated; PH buys parameter accuracy where second-order statistics are blind. Cell is the one candidate for an end-to-end effect [cell-only run].

Before submission

Must have.

- Classical baseline: `spatstat::kppm` (minimum contrast and Palm likelihood) for Thomas, Matérn cluster and LGCP, scored with the same end-to-end score. The other five families have no closed-form fit, which is itself the argument for amortized inference.
- One or two real patterns (candidates in the bank's n range: `redwoodfull`, `finpines`, `spruces`), rescaled to the unit window, scored without an oracle (fit vs CSR vs `kppm`).
- Fig. 1 and the per-model regime curves.
- Confirm the AISTATS 2027 deadline; historically it falls in early October.

Cheap and worth it. Soft routing (simulate from the family mixture weighted by $p(f \mid x)$, using existing posteriors); a calibration check of $p(f \mid x)$.

Future work, framed as the upgrade the factorisation allows. Posterior rather than point estimates for θ (distributional trees or neural posterior estimation) and posterior-predictive sampling; learned permutation-invariant summaries of the raw points (DeepSets / GNN) as the stronger learned baseline.

Anticipated reviewer questions. Why not a classical fit? (§5 baseline.) Why only synthetic data? (real-data demo.) Why point estimates? (limitations; soft routing partly answers it.) Is PH just another feature? (§6: it helps exactly where second-order summaries are blind, and never hurts.) Why is accuracy so low? (§4: the CSR wall; lead with it, do not hide it.)